

A buildable two-board reference

Every panel control is mechanically mounted to the top protoboard. No control or jack floats on a faceplate and there is only one board-to-board connector.

Locked architecture

Use **two matching 24-column x 44-row isolated-pad boards** at 2.54mm pitch. The Top Control Board holds all five pots, both toggles, both buttons, the bicolor LED and four jacks. The Main Board holds Nano R4, SRAM, TL074, power protection, audio and CV circuits.

J1 is a single 1x20 stack connector at C44-V44. Four M3 standoffs at D14, T14, D43 and T43 carry mechanical load. J2 and J3 do not exist in this revision.

Historically matched hardware

Hardware	Required family
Pots	Alpha RD901F-40-00D, 9mm vertical, B10K
Jacks	QingPu WQP-PJ398SM / Thonkiconn style
Buttons	Wurth 430476085716, 12x12mm THT
Quantize	Tayda A-5290, SPDT ON-OFF-ON
Loop/Shot	A-5291-style SPDT ON-ON, same 2.54mm pin pitch
LED	5mm, 3-lead common-cathode red/green

Important protoboard rule

These controls use their real historical footprints. Some holes fall between the stock 2.54mm pads. Drill those lead and locating holes using page 4, then bridge each hardware lead to the named isolated pad on page 5 with the shortest possible tinned solid link. The hardware body remains directly attached to the Top Board.

BOM tied to the coordinate layouts

Reference designators and values match the placement and connection pages.

Qty	Value	Type	Qty	Hardware	Use
4	100K	resistor / PTC	5	Alpha RD901F B10K	P1-P5
1	100mA PTC	resistor / PTC	4	WQP-PJ398SM	audio/CV jacks
10	100n	ceramic or film	2	Wurth 430476085716	Record / Play
8	10K	resistor / PTC	1	Tayda A-5290	Quantize
4	10u	electrolytic	1	A-5291 style ON-ON	Loop / Shot
9	1K	resistor / PTC	1	5mm bicolor common-cathode LED	status
3	1M	resistor / PTC	1	Arduino Nano R4	socketed
2	1N4148	diode	1	23LC1024-I/P + DIP-8 socket	U2
2	1N5817	diode	1	TL074CP + DIP-14 socket	U1
1	1u film	ceramic or film	1	1x20 stack header pair	J1
1	22n	ceramic or film	1	2x5 Eurorack power header	J_PWR
1	250mA PTC	resistor / PTC	2	24x44 isolated-pad boards	minimum hole span 58.42 x 109.22mm
2	2N3904	transistor	4	M3 10-12mm standoffs	board stack
1	39K	resistor / PTC			
1	4.7u	electrolytic			
2	BAT85	diode			
1	VIN ENABLE	jumper			

The only inter-board connector

J1 is one straight 1x20 header. Pin 1 is C44 on both boards and pin 20 is V44.

Pin	Hole	Net
1	C44	+5N
2	D44	GND
3	E44	HEAD1
4	F44	HEAD2
5	G44	HEAD3
6	H44	AGE
7	I44	MIX
8	J44	REC_BTN
9	K44	PLAY_BTN
10	L44	Q_OCT
11	M44	Q_SEMI
12	N44	PLAY_MODE
13	O44	LED_R_OUT
14	P44	LED_G_OUT
15	Q44	AUDIO_IN_TIP
16	R44	AUDIO_OUT_TIP
17	S44	REC_TIP
18	T44	PLAY_TIP
19	U44	GND
20	V44	GND

Stacking

Fit a 1x20 female socket to the Main Board and the matching long-pin male header to the Top Board. Mark pin 1 at C44 in red on both boards before soldering.

Fit M3 standoffs at D14, T14, D43 and T43. The header carries signals only; the standoffs resist control and patch-cable forces.

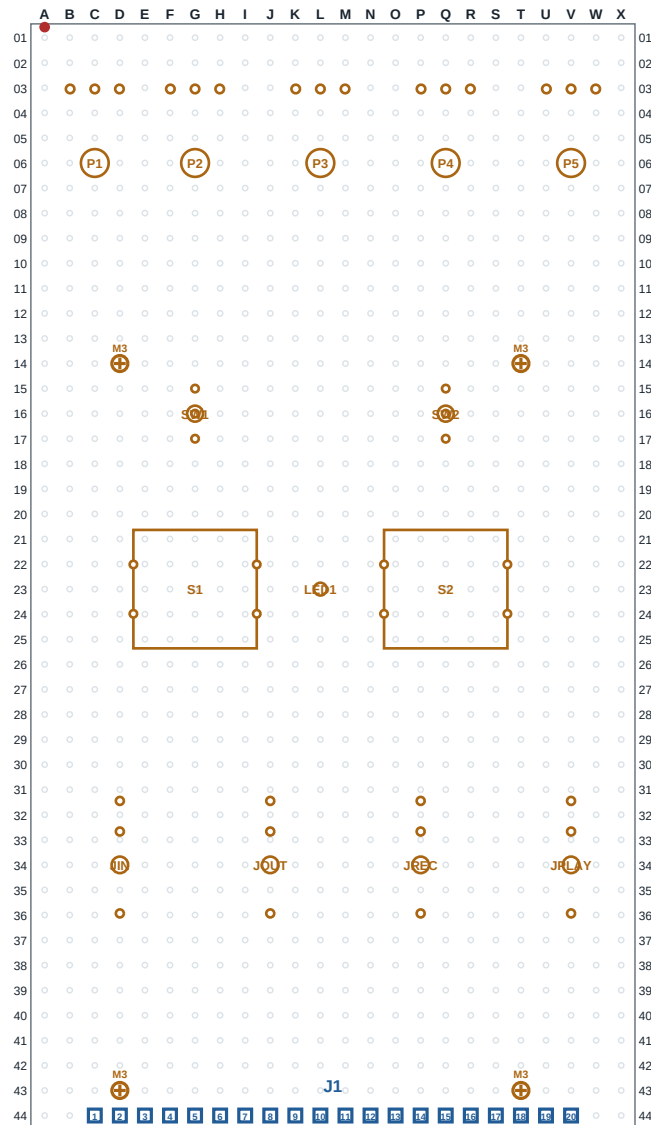
No panel harness

Pots, switches, buttons, LED and jacks terminate on the Top Board. Their shared +5V and ground buses also live there. Nothing connects through loose J2/J3 wiring.

J_PWR remains on the Main Board so rack power never travels through the control hardware board.

Top Control Board - real hardware footprints

Component side. Gold outlines show the actual historical hardware bodies and lead patterns; blue squares are J1.



Direct mounting

RD901F pots are rotated 90 degrees so their 2.50mm electrical-pin row sits above each shaft. WQP-PJ398SM jacks use sleeve / switch / tip offsets -6.48 / -3.38 / +4.92mm from the shaft. Leave jack switch-normal pins unconnected. Wurth buttons use all four mechanical leads, one electrical side per net.

Top Control Board - lead-to-pad bridges

After drilling the exact footprints, bridge each listed hardware lead to the named isolated pad with a short rigid tinned lead.

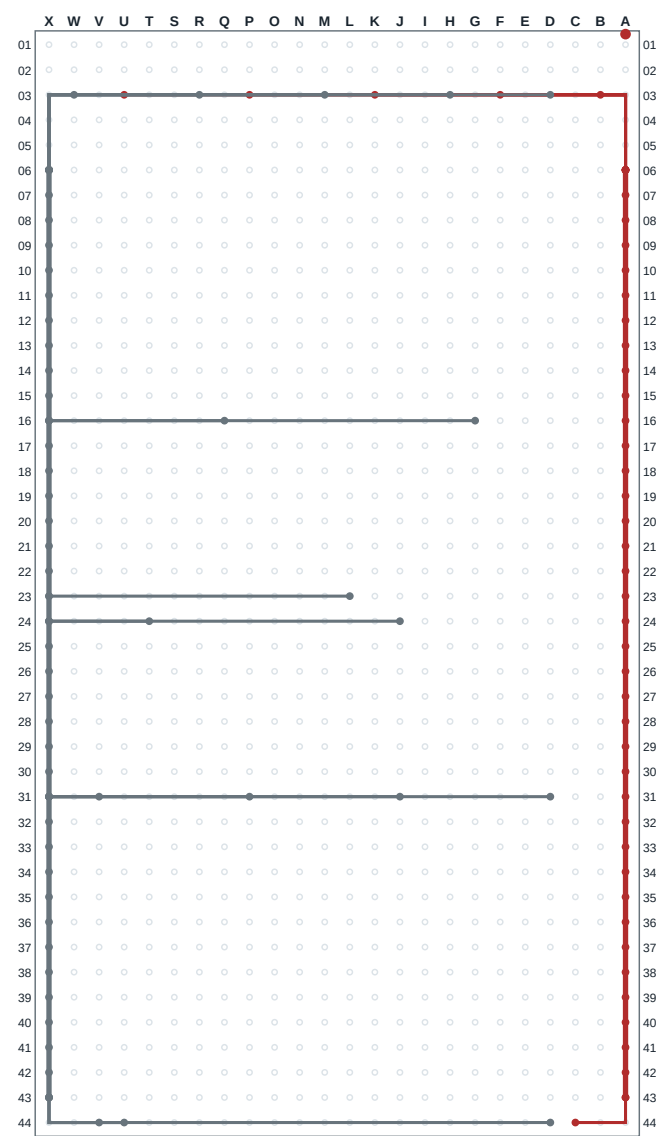
Ref	Historical part	Named grid-pad bridge
P1	HEAD 1 / RD901F	B3=+5N; C3=HEAD1; D3=GND
P2	HEAD 2 / RD901F	F3=+5N; G3=HEAD2; H3=GND
P3	HEAD 3 / RD901F	K3=+5N; L3=HEAD3; M3=GND
P4	AGE / RD901F	P3=+5N; Q3=AGE; R3=GND
P5	MIX / RD901F	U3=+5N; V3=MIX; W3=GND
SW1	A-5290 QUANTIZE	G15=Q_OCT; G16=GND; G17=Q_SEMI
SW2	A-5291 LOOP/SHOT	Q15=PLAY_MODE; Q16=GND
S1	WURTH RECORD	E22=REC_BTN; J24=GND
S2	WURTH PLAY	O22=PLAY_BTN; T24=GND
LED1	5MM RED/GREEN	K23=LED_R_OUT; L23=GND; M23=LED_G_OUT
JIN	PJ398SM AUDIO IN	D31=GND; D36=AUDIO_IN_TIP
JOUT	PJ398SM AUDIO OUT	J31=GND; J36=AUDIO_OUT_TIP
JREC	PJ398SM RECORD CV	P31=GND; P36=REC_TIP
JPLAY	PJ398SM PLAY CV	V31=GND; V36=PLAY_TIP

Orientation rules

Pot pins are +5V / wiper / GND in the order printed above; reverse the two outer connections only if the physical rotation is opposite. SW1 centre is GND and its two throws are octave/semitone. SW2 common is GND and its selected throw is PLAY_MODE. LED centre lead is common cathode GND.

Top solder pass 1 - shared +5V and ground

Solder side, mirrored. The A-column +5V and X-column ground spines eliminate repeated harness returns.



ID	Endpoints	Net
T001	A6..A43 ALL	+5N
T002	B3-A6	+5N
T003	F3-A6	+5N
T004	K3-A6	+5N
T005	P3-A6	+5N
T006	U3-A6	+5N
T007	C44-A43	+5N
T011	X6..X43 ALL	GND
T012	D3-X6	GND
T013	H3-X6	GND
T014	M3-X6	GND
T015	R3-X6	GND
T016	W3-X6	GND

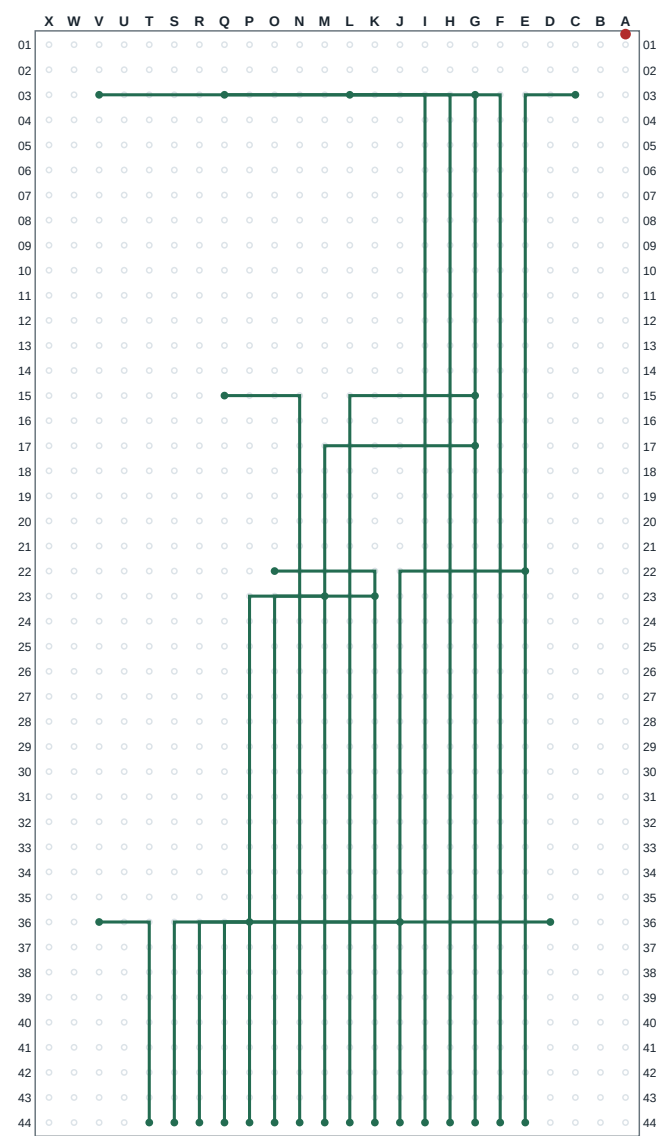
ID	Endpoints	Net
T017	G16-X16	GND
T018	Q16-X16	GND
T019	J24-X24	GND
T020	T24-X24	GND
T021	L23-X23	GND
T022	D31-X31	GND
T023	J31-X31	GND
T024	P31-X31	GND
T025	V31-X31	GND
T026	D44-X43	GND
T027	U44-X43	GND
T028	V44-X43	GND

Meter before J1

All pot high sides reach +5N. All pot low sides, jack sleeves, button returns, toggle commons and LED cathode reach GND. +5N and GND must remain open to each other.

Top wiring - controls and jack tips

Solder side, mirrored. Each green route terminates at exactly one J1 signal pin.



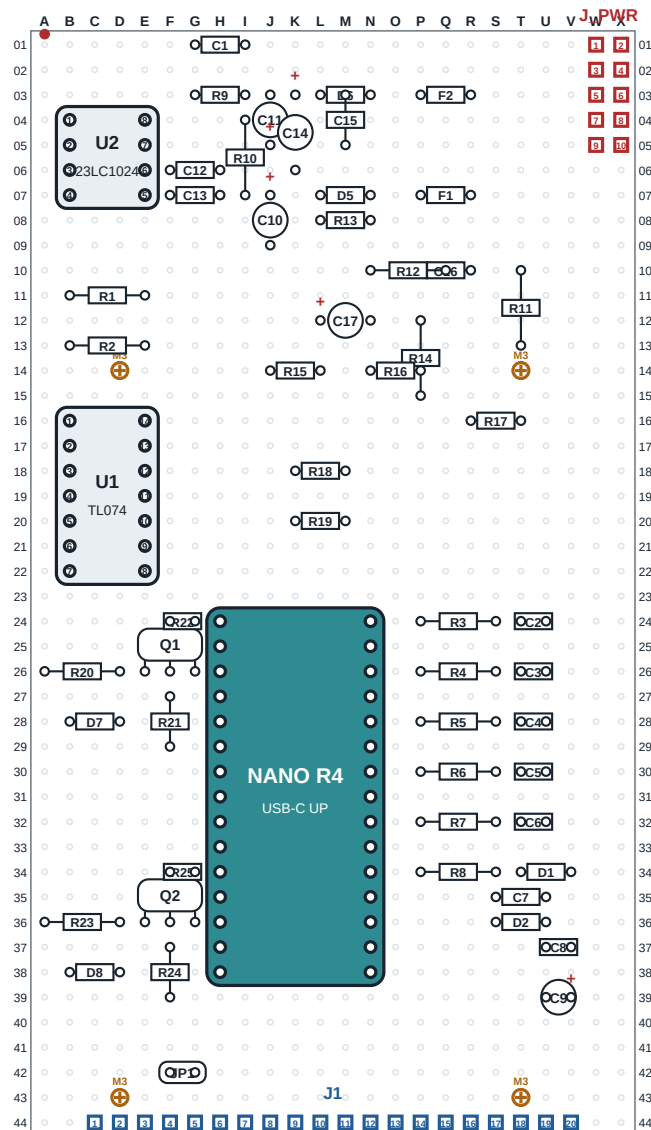
ID	Endpoints	Net	ID	Endpoints	Net
T008	Q3-H44	AGE	T034	V3-I44	MIX
T009	D36-Q44	AUDIO_IN_TIP	T035	O22-K44	PLAY_BTN
T010	J36-R44	AUDIO_OUT_TIP	T036	Q15-N44	PLAY_MODE
T029	C3-E44	HEAD1	T037	V36-T44	PLAY_TIP
T030	G3-F44	HEAD2	T038	G15-L44	Q_OCT
T031	L3-G44	HEAD3	T039	G17-M44	Q_SEMI
T032	M23-P44	LED_G_OUT	T040	E22-J44	REC_BTN
T033	K23-O44	LED_R_OUT	T041	P36-S44	REC_TIP

No floating controls

Every control body is soldered or mechanically fixed to this board. The only short links are the rigid lead-to-pad bridges documented on page 5; there is no faceplate harness.

Main Board - complete component placement

Component side. All active electronics and Eurorack power remain on this lower board.



Orientation

Nano USB-C faces row 23. U2 notch and U1 notch face row 1. J_PWR red stripe is pins 1/2 at W1/X1. J1 pin 1 is C44. Enlarge D14/T14/D43/T43 to 3.2mm only after continuity testing the surrounding pads.

Main Board - exact coordinates, part 1

These lead holes are authoritative when labels crowd the placement drawing.

Ref	Value	Lead holes	Nets
R1	10K	B11 / E11	SIO2 / +5N
R2	10K	B13 / E13	HOLD / +5N
C1	100n	G1 / I1	+5N / GND
JP1	VIN ENABLE	F42 / G42	+12P / VIN
R3	1K	P24 / S24	HEAD1 / HEAD1_F
C2	100n	T24 / U24	HEAD1_F / GND
R4	1K	P26 / S26	HEAD2 / HEAD2_F
C3	100n	T26 / U26	HEAD2_F / GND
R5	1K	P28 / S28	HEAD3 / HEAD3_F
C4	100n	T28 / U28	HEAD3_F / GND
R6	1K	P30 / S30	AGE / AGE_F
C5	100n	T30 / U30	AGE_F / GND
R7	1K	P32 / S32	MIX / MIX_F
C6	100n	T32 / U32	MIX_F / GND
R8	1K	P34 / S34	AUDIO_ADC / AUDIO_FILT
C7	22n	S35 / U35	AUDIO_FILT / GND
D1	BAT85	T34 / V34	AUDIO_FILT / +5N
D2	BAT85	U36 / S36	GND / AUDIO_FILT
C8	100n	U37 / V37	GND / +5N
C9	10u	V39 / U39	+5N / GND
F1	250mA PTC	R7 / P7	+12_BUS / +12_FUSED
D5	1N5817	N7 / L7	+12_FUSED / +12P
F2	100mA PTC	R3 / P3	-12_BUS / -12_FUSED
D6	1N5817	N3 / L3	-12_FUSED / -12P
C10	10u	J7 / J9	+12P / GND
C11	10u	J3 / J5	-12P / GND
C12	100n	F6 / H6	+12P / GND

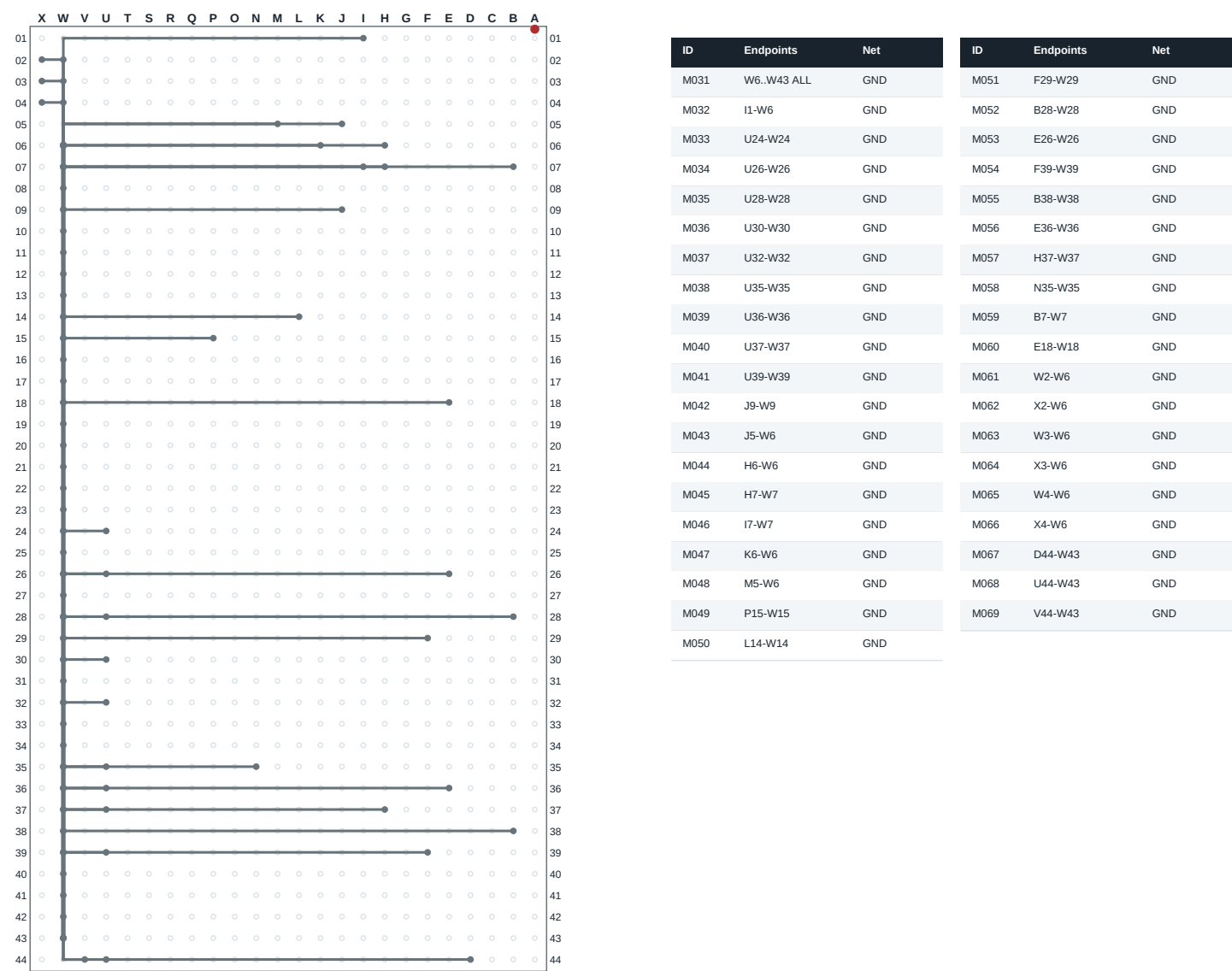
Main Board - exact coordinates, part 2

These lead holes are authoritative when labels crowd the placement drawing.

Ref	Value	Lead holes	Nets
C13	100n	F7 / H7	-12P / GND
R9	10K	G3 / I3	+5N / VREF_DIV
R10	10K	I4 / I7	VREF_DIV / GND
C14	10u	K3 / K6	VREF_DIV / GND
C15	100n	M3 / M5	VREF_DIV / GND
C16	1u film	P10 / R10	AUDIO_IN_TIP / AIN_BIASED
R11	1M	T10 / T13	AIN_BIASED / VREF
R12	100K	Q10 / N10	AIN_BIASED / AIN_SUM
R13	39K	L8 / N8	AIN_SUM / AUDIO_ADC
C17	4.7u	L12 / N12	DAC_RAW / AOUT_AC
R14	100K	P12 / P15	AOUT_AC / GND
R15	10K	J14 / L14	AOUT_FB / GND
R16	10K	N14 / P14	AOUT_FB / AOUT_OP
R17	1K	R16 / T16	AOUT_OP / AUDIO_OUT_TIP
R18	1K	K18 / M18	LED_R_IN / LED_R_OUT
R19	1K	K20 / M20	LED_G_IN / LED_G_OUT
R20	100K	A26 / D26	REC_TIP / REC_BASE
R21	1M	F27 / F29	REC_BASE / GND
D7	1N4148	B28 / D28	GND / REC_BASE
Q1	2N3904	E26 / F26 / G26	GND / REC_BASE / REC_CV
R22	10K	F24 / G24	REC_CV / +5N
R23	100K	A36 / D36	PLAY_TIP / PLAY_BASE
R24	1M	F37 / F39	PLAY_BASE / GND
D8	1N4148	B38 / D38	GND / PLAY_BASE
Q2	2N3904	E36 / F36 / G36	GND / PLAY_BASE / PLAY_CV
R25	10K	F34 / G34	PLAY_CV / +5N

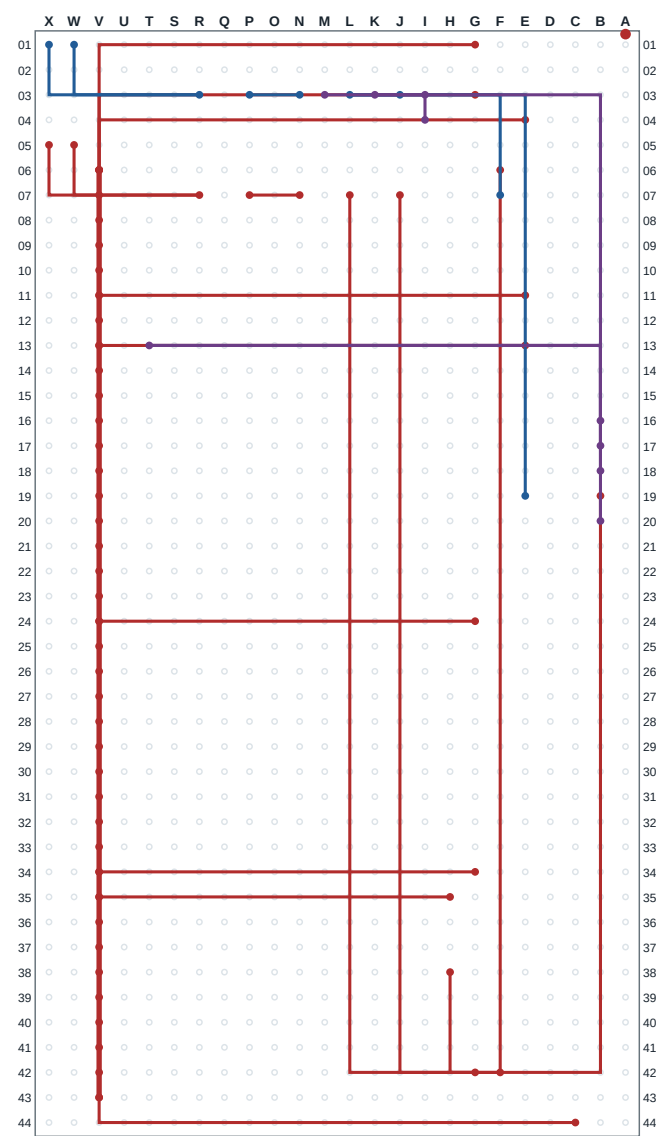
Main solder pass 1 - ground

Solder side, mirrored. W6-W43 is the continuous bare ground spine; all other ground links are insulated.



Main wiring - power and reference

Solder side, mirrored. Check protected rails before inserting Nano, SRAM or TL074.



ID	Endpoints	Net
M001	F42-L7-J7-F6-B19	+12P
M002	R7-W5-X5	+12_BUS
M003	P7-N7	+12_FUSED
M004	V6-.V43 ALL	+5N
M005	E11-V11	+5N
M006	E13-V13	+5N
M007	G1-V6	+5N
M008	G3-V6	+5N
M009	G24-V24	+5N
M010	G34-V34	+5N

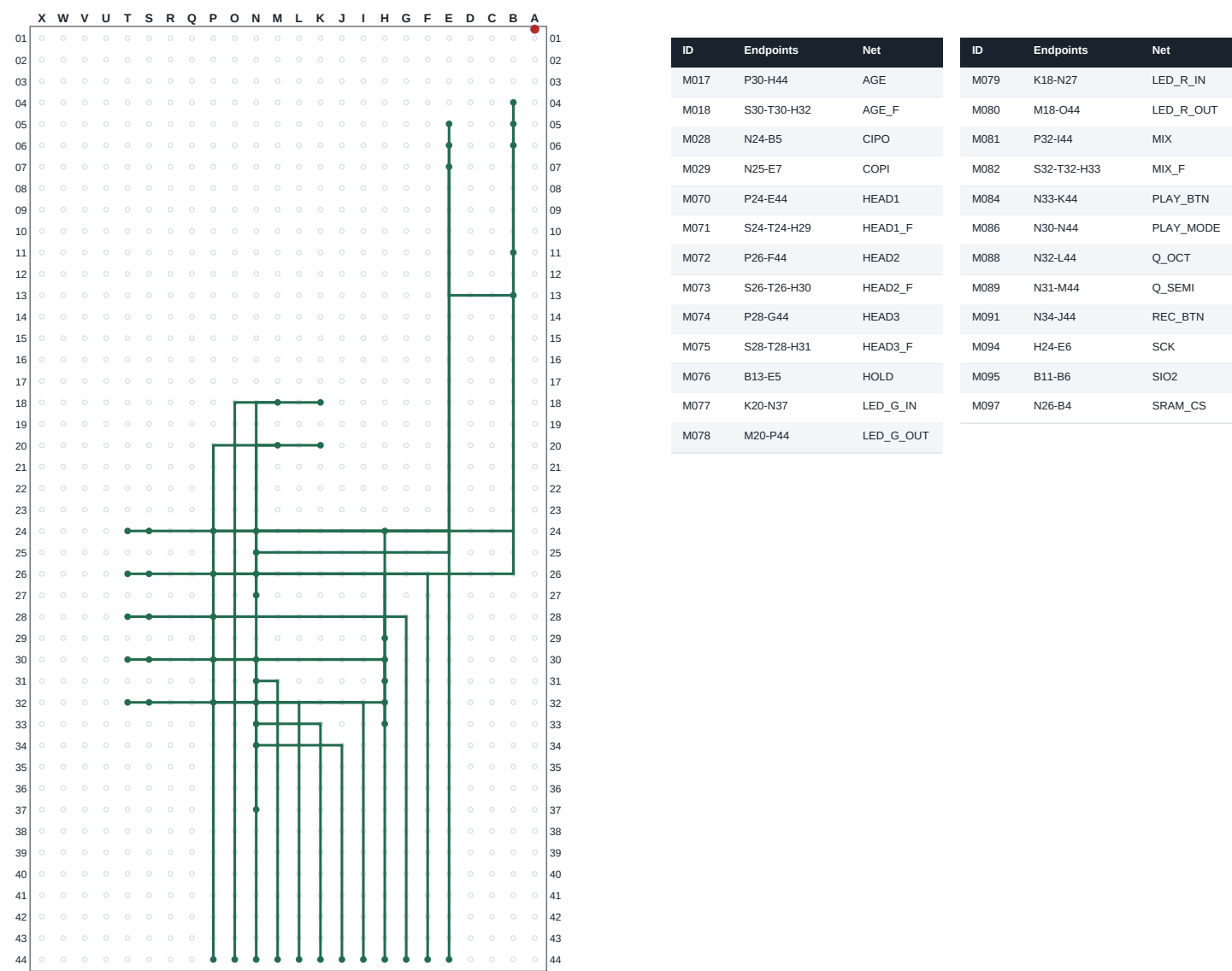
ID	Endpoints	Net
M011	H35-V35	+5N
M012	E4-V6	+5N
M013	C44-V43	+5N
M014	L3-J3-F7-E19	-12P
M015	R3-W1-X1	-12_BUS
M016	P3-N3	-12_FUSED
M098	G42-H38	VIN
M099	T13-B16-B17-B20	VREF
M100	I3-I4-K3-M3-B18	VREF_DIV

Expected rails

Before ICs: +12P after F1/D5, -12P after F2/D6, and no rail-to-ground short. After Nano provides +5N, VREF and VREF_DIV should sit near 2.5V.

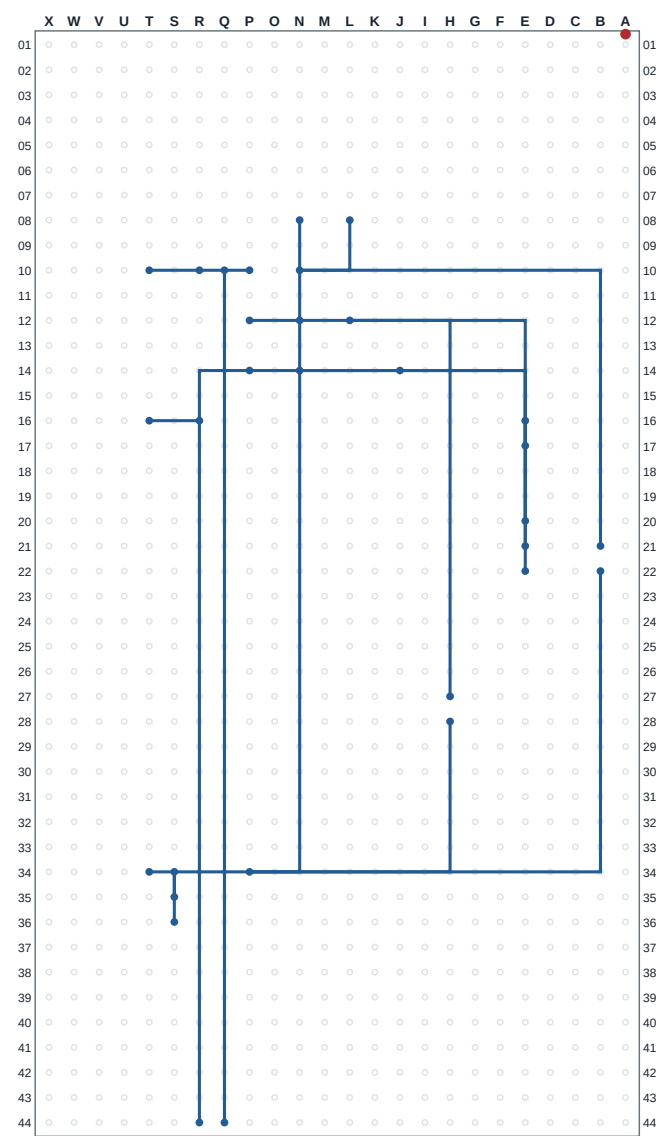
Main solder pass 3 - SRAM and controls

Solder side, mirrored. These are the SPI, five pot, buttons, toggles and LED routes.



Main wiring - audio path

Solder side, mirrored. Purple routes carry VREF; blue routes carry audio and DAC signals.



ID	Endpoints	Net
M019	R10-T10-Q10	AIN_BIASED
M020	N10-L8-B21	AIN_SUM
M021	N12-P12-E20	AOUT_AC
M022	J14-N14-E21	AOUT_FB
M023	P14-R16-E22	AOUT_OP
M024	P34-N8-B22	AUDIO_ADC

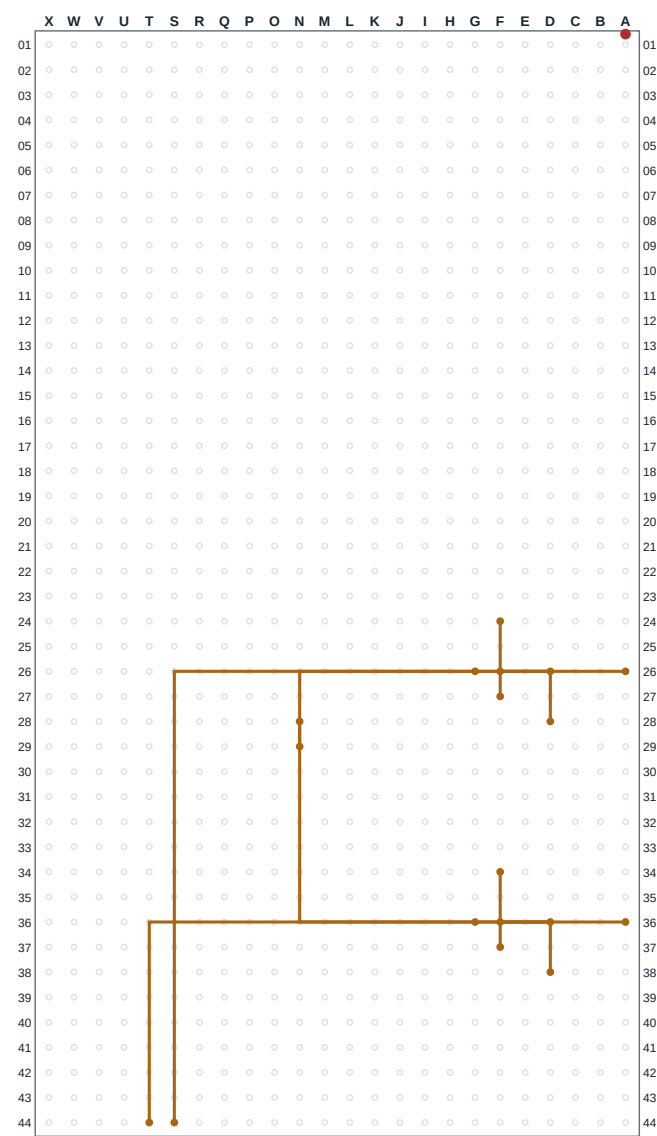
ID	Endpoints	Net
M025	S34-S35-T34-S36-H28	AUDIO_FILT
M026	P10-Q44	AUDIO_IN_TIP
M027	T16-R44	AUDIO_OUT_TIP
M030	L12-H27	DAC_RAW
M096	E17-E16	SPARE_LOOP

DC check

No input: U1 pin 7 and Nano A1 should rest near 2.5V.
Audio-output tip should rest near 0V DC after the output coupling stage.

Main solder pass 5 - Record and Play CV

Solder side, mirrored. Two identical NPN stages turn positive Eurorack pulses into protected active-low inputs.



ID	Endpoints	Net	ID	Endpoints	Net
M083	D36-F37-D38-F36	PLAY_BASE	M090	D26-F27-D28-F26	REC_BASE
M085	G36-F34-N28	PLAY_CV	M092	G26-F24-N29	REC_CV
M087	A36-T44	PLAY_TIP	M093	A26-S44	REC_TIP

Transistor check

Verify the actual 2N3904 E-B-C lead order before insertion.
D7/D8 anodes reach GND and bands face the base nodes.
Collectors idle HIGH through R22/R25 and pulse LOW on a valid positive trigger.

Assembly and staged power-up

Do not insert Nano, U1 or U2 until the passive continuity and rail tests pass.

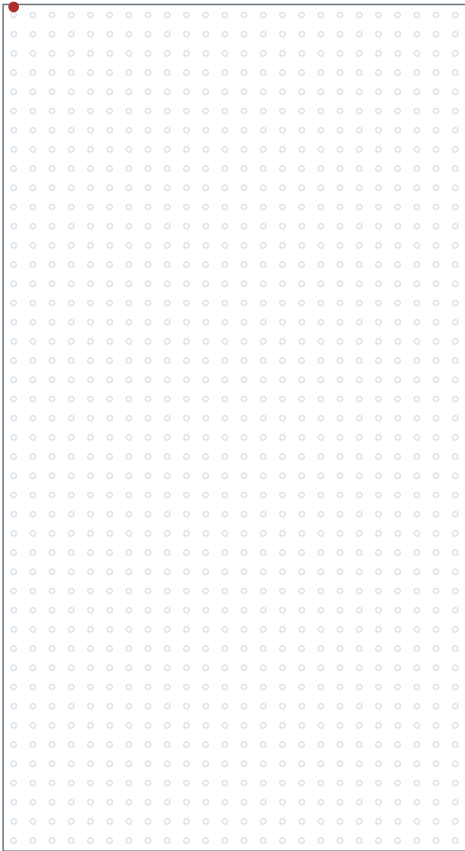
Stage	Stop point
1	Cut two 24x44 boards. Mark A1, J1 pin 1 and all four standoff holes.
2	Drill Top Board hardware footprints from page 4; dry-fit every exact historical part.
3	Fit top hardware and make the short lead-to-pad bridges from page 5.
4	Complete Top Board bus and signal passes; meter J1 pins against page 3.
5	Fit Main Board sockets, parts and J_PWR; leave Nano/U1/U2 out.
6	Complete Main solder passes in order and meter every rail.
7	Power Main Board current-limited. Confirm +12P, -12P, +5N and 2.5V VREF.
8	Fit U1, then Nano, then U2. Test each stage before stacking.
9	Install four standoffs, align J1 C44 on both boards, then stack.
10	Run diagnostic firmware before audio firmware.

USB rule

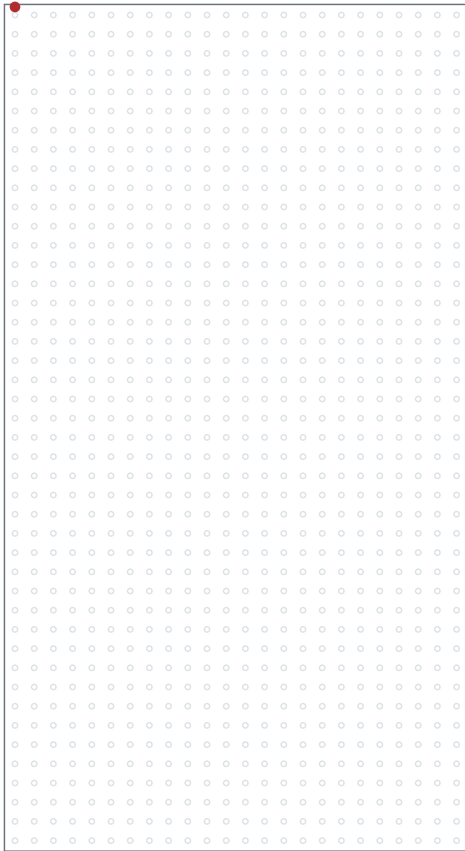
For USB-only flashing, remove the module from the rack, open JP1 and separate J1. Never let USB back-power the unpowered TL074/audio board through signal clamps. Reconnect J1 only after USB is removed or the rack is intentionally powered under the documented test condition.

Actual-size board grids and build record

Print at 100%. Both grids below are true 2.54mm pitch; all earlier placement drawings are enlarged.



TOP CONTROL BOARD



MAIN BOARD

The line below must measure exactly 50mm.



Build record

Builder: _____ Date: _____ Board revision: _____

+12P: _____V -12P: _____V +5N: _____V VREF: _____V

Firmware file / commit: _____

SRAM: PASS / FAIL Audio: PASS / FAIL Controls/CV: PASS / FAIL